

Mathematics Internal Assessment:

Determining the Arc Length of a

Rollercoaster Loop

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Introduction

Every time I go to the amusement park, I have always been attracted to the rollercoasters. I have always wondered how the unique shapes that the roller coaster tracks were made, especially the loop, where a rider does a 360 turn.

Most of the time, roller coasters are the main attraction of any amusement park. To make it more dopamine inducing for the visitors, designers often create loops inside the rollercoaster system. Originally, the vertical loop used a completely circular design, but due to safety issues associated with high g forces, the modern roller coaster loop is now an inverted tear drop shape, reducing the danger of neck injuries. As this shape is considerably harder to create and design, roller coaster designers would have to use a more complex way to model these loops compared to a simple circle, so there would need to be a way to find out the length and size of the loop, so that materials can be prepared and to make sure that the roller coaster loop stays structurally stable and not fall in operation.



Figure 1: A rollercoaster from a local amusement park (Alvey & Alvey)

As such, I wanted to investigate more about how these safety considerations had affected the shape, length, and other parameters of the loop.

Methodology

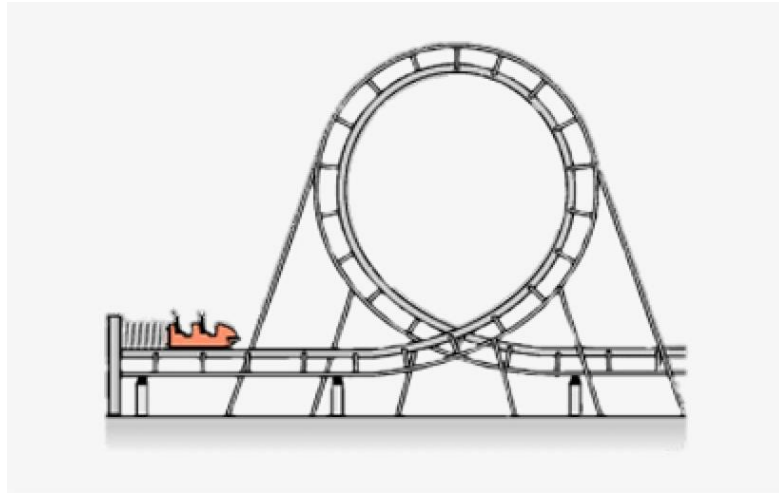


Figure 2: A side view of a rollercoaster loop (Circus Clipart Roller Coaster - Forces Roller Coaster Loop)

Firstly, I would use calculus to determine a mathematical expression that would be used to determine the curve, then using that expression, the characteristics of the loop would be determined.

As this investigation would include the calculation of forces and motion, integration and kinematics would be utilised. Trigonometry and geometry are also utilised as the object investigated is a real-life object that is curved.

Calculation

List of nomenclature

Variable	Meaning
s	Arc length of a small section of a track (m)
θ	Angle of a circular arc (degrees)
r	Radius (m)
κ	Curvature
q	Specific constant of proportionality
n	Number of divided intervals for a curve
P_i	A point on a curve
i	A specific interval for a curve
L	Arc length of a clothoid curve (m)
L_T	Total arc length of the rollercoaster (m)
w	Rate of change of z

Figure 3: A list of variables used

Determining the track and the model of the loop

To start, after looking at a rollercoaster loop, as the shape is unique, I wanted to break it apart into different sections so that calculations could be done in an easier manner. For the early stages, I would only the rollercoaster loop track is then divided into equidistant segments, making it easier to calculate the length of the track using geometry.

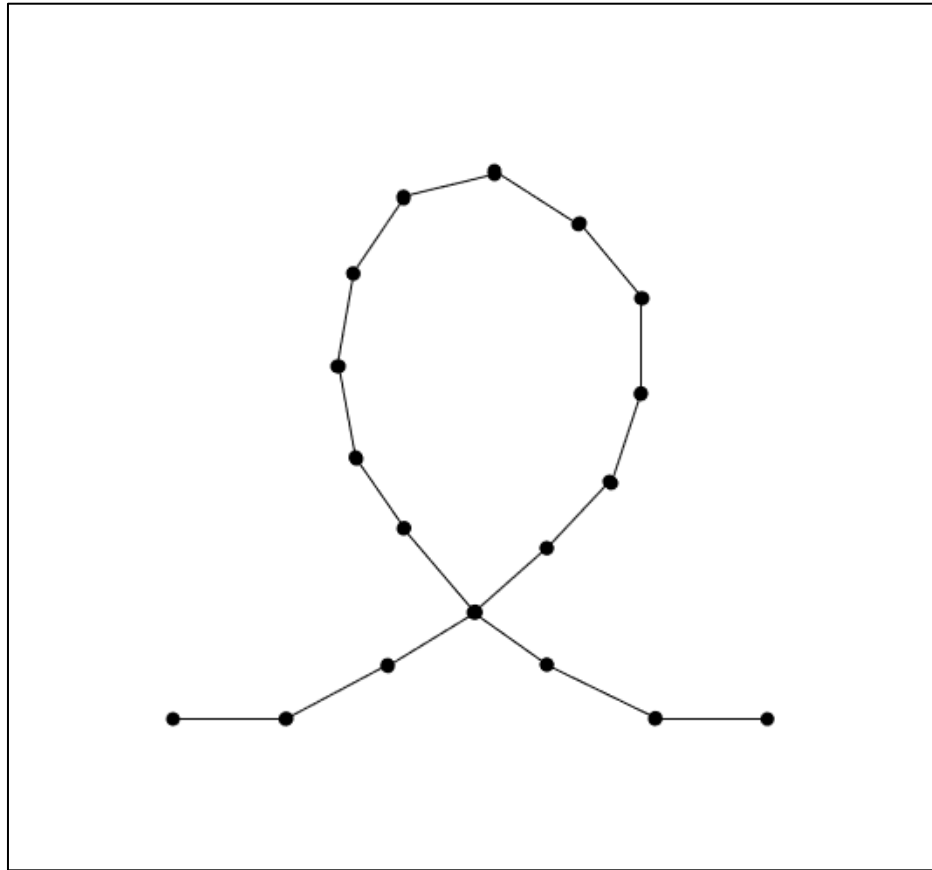


Figure 4: A rollercoaster loop divided into different sections

We can consider an exceedingly small section of a track of the loop with arc length Δs . As the length of the track is super small, the radius can be considered as constant over its length as its curvature is constant. As such, we can make it become a circular arc with an internal angle of $\Delta\theta$.

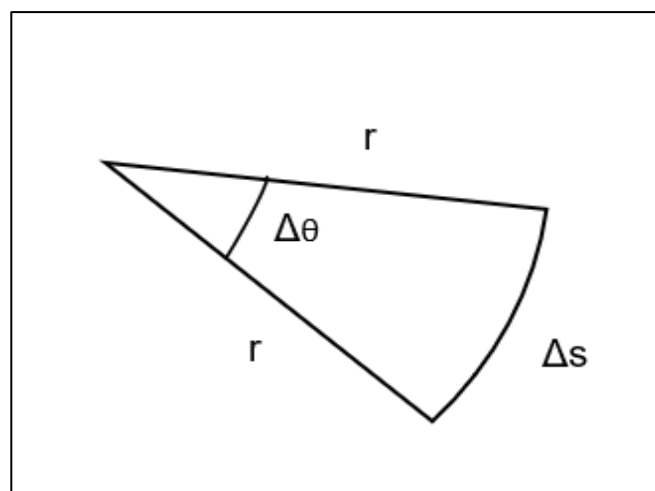


Figure 5: Diagram of the Circular Arc

Then, we can use the arc length formula

$$\Delta s = r \times \Delta\theta \quad (1)$$

Which can be changed to

$$\frac{\Delta\theta}{\Delta s} = \frac{1}{r} \quad (2)$$

The horizontal component of the circular arc is denoted as Δx and the vertical component is denoted as Δy .

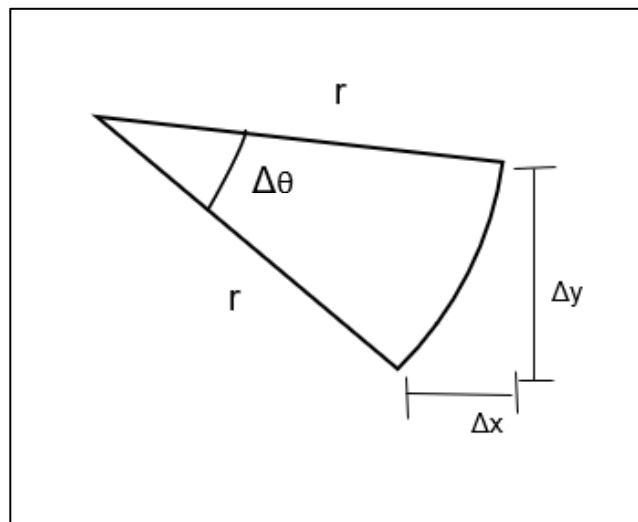


Figure 6: Circular arc with horizontal and vertical component

The arc length of the circular arc can be considered to become a triangle, and so using trigonometry, the two components can be related to trigonometric angles:

$$\frac{\Delta x}{\Delta s} = \cos(\theta) \quad (3)$$

$$\frac{\Delta y}{\Delta s} = \sin(\theta) \quad (4)$$

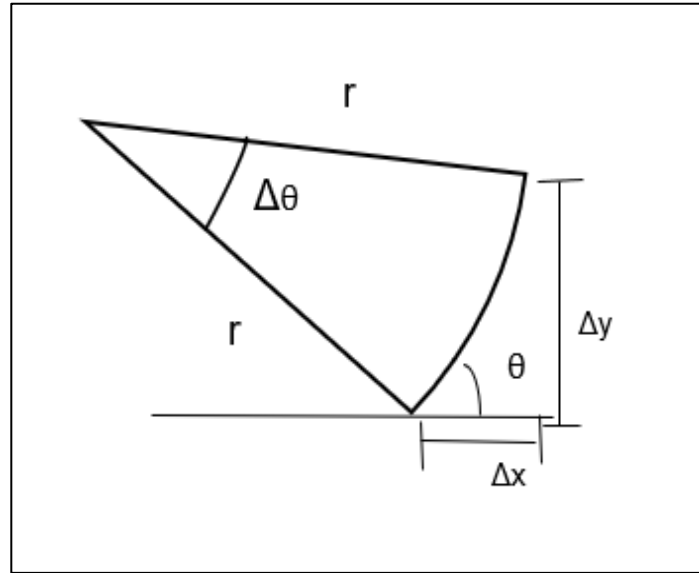


Figure 7: Triangle in the exterior of the arc

As the arc length can be considered as infinitesimally small, we can find the differential equations for formulas (2), (3), (4) by taking its limits:

$$\lim_{\Delta s \rightarrow 0} \frac{\Delta \theta}{\Delta s} = \frac{d\theta}{ds} = \frac{1}{r} \quad (5)$$

$$\lim_{\Delta s \rightarrow 0} \frac{\Delta x}{\Delta s} = \frac{dx}{ds} = \cos(\theta) \quad (6)$$

$$\lim_{\Delta s \rightarrow 0} \frac{\Delta y}{\Delta s} = \frac{dy}{ds} = \sin(\theta) \quad (7)$$

After finding out the rate of change for the vertical component and the horizontal component, I was stuck on what to do to find out the total arc length of the rollercoaster loop. I knew that I needed to relate equation (5) to (6) and (7) somehow, so after some research, I found out that the curvature can be used to link all the equations together.

Curvature

The curvature κ is defined as the rate of change of the angle through which the tangent to a curve turns in moving along the curve and which for a circle is equal to the reciprocal of the radius. (*Curvature definition & meaning*)

$$\kappa = \frac{1}{r} \quad (8) \quad (\text{Radius of curvature formula - learn formula for radius of curvature})$$

The radius decreases at a constant rate with respect to the arclength; so, the curvature would increase at a constant rate with respect to the arc length. As the curvature is proportional to the arc length, a formula can be derived from this relationship, which is

$$\kappa = qs \quad (9)$$

Where q is the specific constant of proportionality. The constant of proportionality is the constant value of the ratio between two proportional quantities. (*Constant of proportionality with example*) Formula (5) would equal to

$$\frac{d\theta}{ds} = \kappa \quad (10)$$

After this, integration would be done after changing the formula to one with respect to θ .

$$\theta = \int \kappa \, ds \quad (11)$$

Substituting (9) into (11), θ becomes

$$\theta = \int q s \, ds$$

$$\theta = \frac{1}{2}qs^2 \quad (12)$$

Then, we can substitute (12) into (6) and (7), becoming

$$\frac{dx}{ds} = \cos\left(\frac{1}{2}qs^2\right) \quad (13)$$

$$\frac{dy}{ds} = \sin\left(\frac{1}{2}qs^2\right) \quad (14)$$

Then, we can change formula (13) and (14) to be integrated:

$$x = f(s) = \int \cos\left(\frac{1}{2}qs^2\right)(15)$$

$$y = g(s) = \int \sin\left(\frac{1}{2}qs^2\right)(16)$$

And can then make a vector function

$$\vec{h}(t) = \left\langle \int_a^b \cos\left(\frac{1}{2}qs^2\right), \int_a^b \sin\left(\frac{1}{2}qs^2\right) \right\rangle (17)$$

These functions give a graph

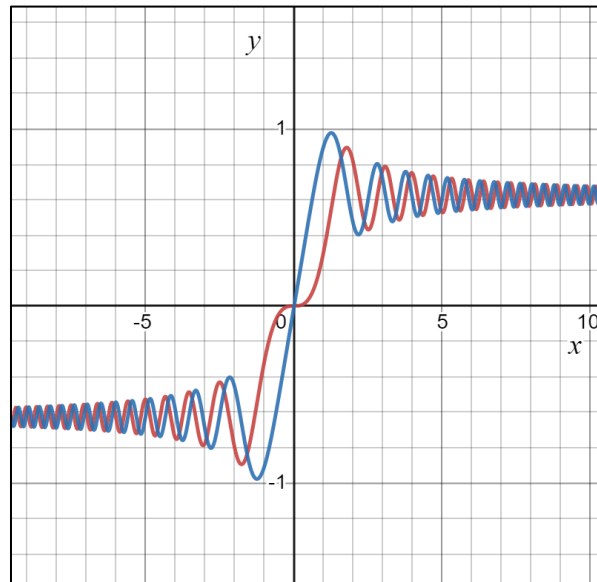


Figure 8: Graph of the x function and y function

Where the purple curve is (15) and the red curve is (16).

After finding the vector functions for x and y , there was still a problem with my investigation; the graph does not really look like a rollercoaster track at all. I needed to relate formula (15) and (16) in a single graph so that I could visualise and find out the arc length of a rollercoaster loop. As such, a parametric curve is created to find out the arc length.

Parametric curves

We can use formulae (15) and (16) and create a parametric equation for the roller coaster in two dimensions as it goes through a vertical loop. A parametric equation defines the coordinate of a point in x and y with another variable, called a parameter. For example, the normal function for a unit circle is

$$x^2 + y^2 = 1 \quad (18)$$

Which graphs out

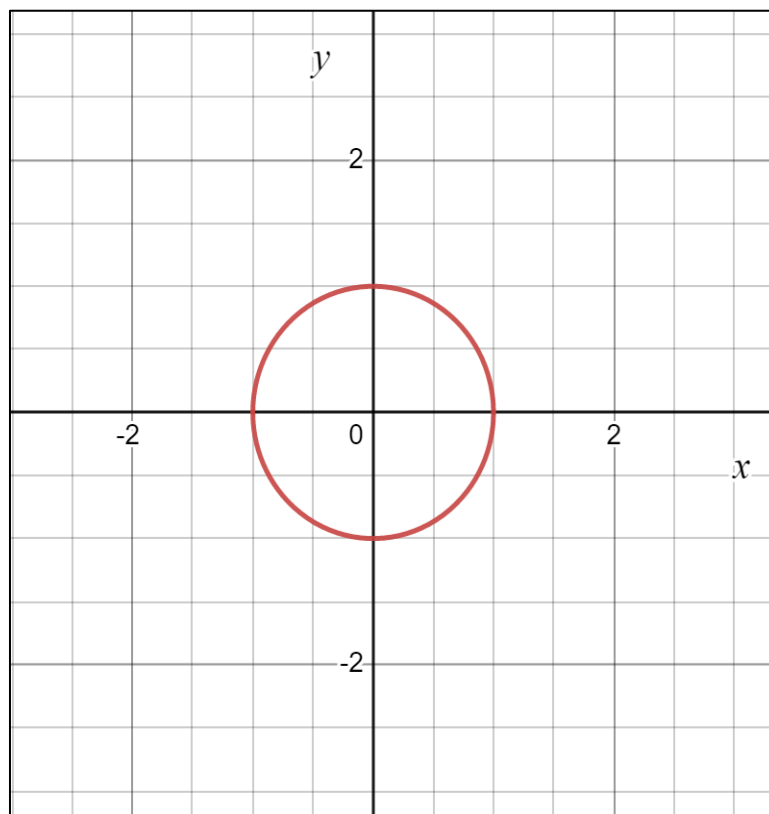


Figure 9: Graph of a unit circle using a standard function

The circle can also be drawn out using parametric equations for $0 \leq t \leq 2\pi$

$$x = \cos(t)$$

$$y = \sin(t)$$

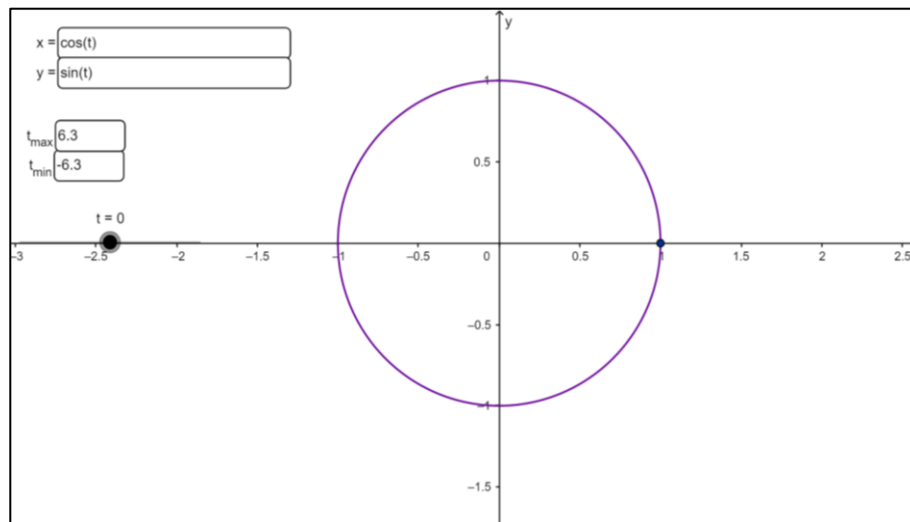


Figure 10: Unit circle using parametric equations

In this case, plotting (15) and (16) will result in a curve that resembles a clothoid curve, or Euler's spiral.

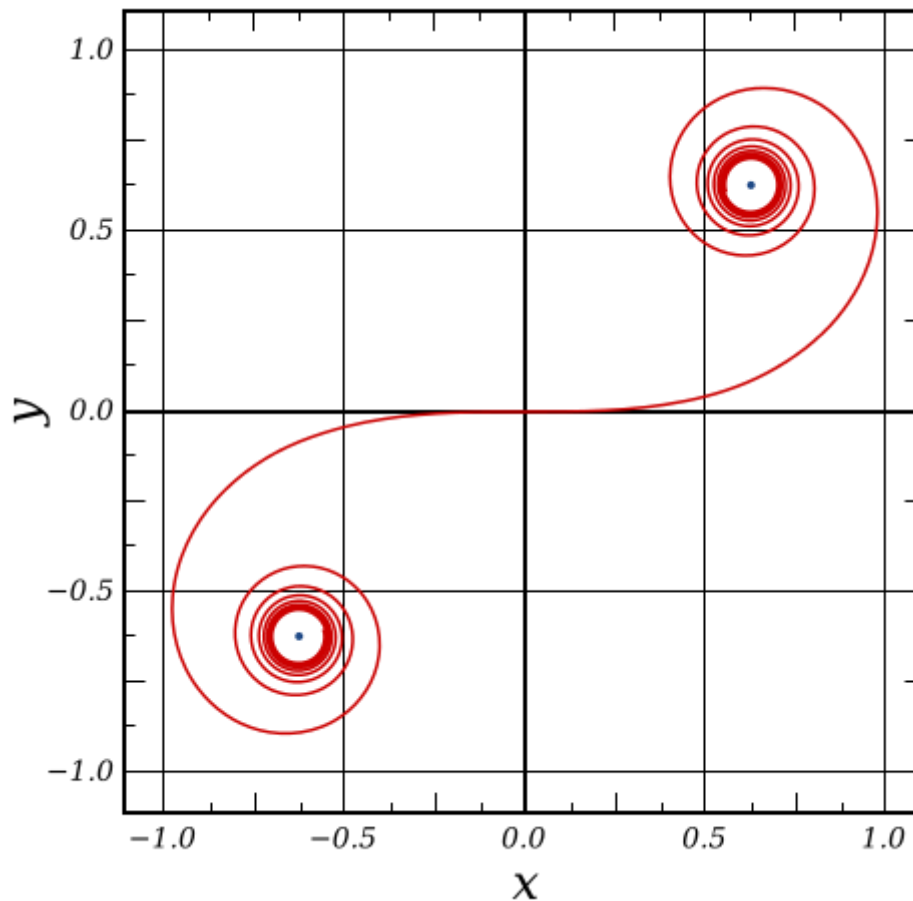


Figure 11: Equation (15) and (16) plotted as parametric equations (File:Cornu Spiral.svg)

Learning about how trigonometric functions can be utilised to create interesting functions was a unexpected surprise. Originally, I thought that the parametric equations can only be used to describe a vector but learning that different functions can be inputted to create graphs is a unique experience for me.

Euler's Spiral

The clothoid curve's curvature has a positive linear relationship with the arc length. It forms a double spiral that has a rotational symmetry of two orders. The curve starts at the origin and starts to curve around its curvature increases with the arc length. Eventually, it converges around an inflection point. We will define the minimum of the vertical loop as the y axis, and the start of the loop at the left side, so we will only look at the quadrant (I) for the cartesian coordinate system.

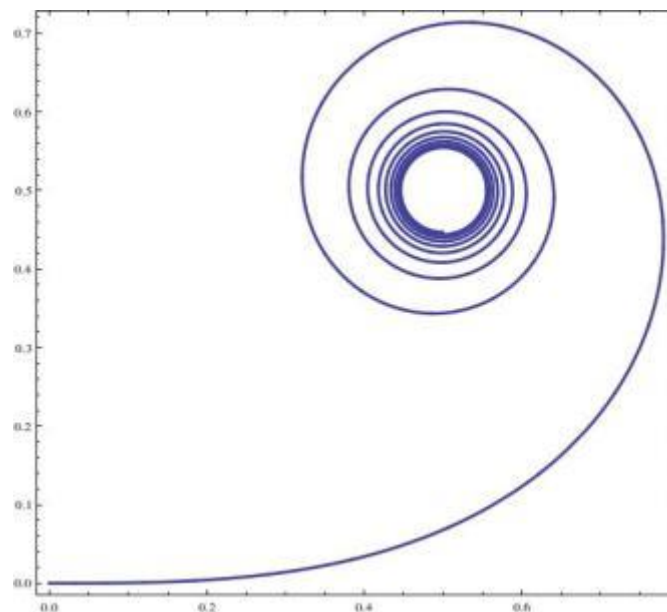


Figure 12: Graphical Representation of Euler's Spiral (Lynch)

However, the clothoid curve does not look like a roller coaster loop at this moment. This is because roller coaster designers only use a section of the graph to design a vertical loop for the roller coaster. A roller coaster loop looks like this:

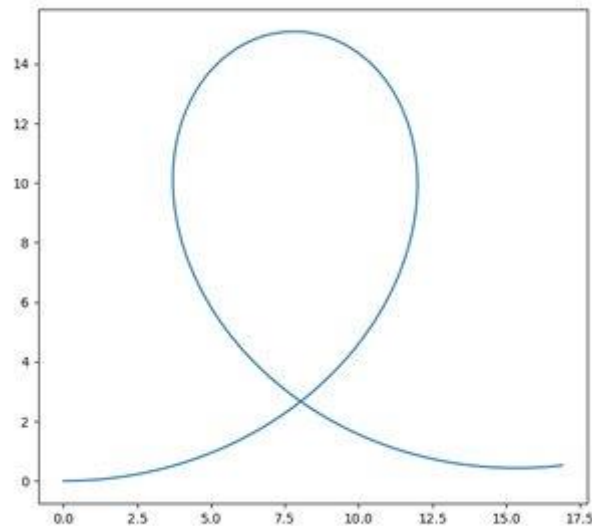


Figure 13: Graphical representation of a rollercoaster loop (Rashid et al.)

To use the clothoid curve, I will use the curve up until the maximum point of the curve, which can determine the first half of the vertical loop of the roller coaster. The other half can be determined by reflecting the curve as they are symmetric.

Finding out how to graphically represent a rollercoaster loop was a breakthrough, but I still needed a way to relate the function for the rollercoaster loop to its length. I did not know if utilising the magnitude of the vector is equivalent to the arc length of a curve. In addition, as the Euler Spiral has trigonometric functions, this may affect the accuracy of using magnitude for the arc length. As such, I needed to prove that the formula for the vector's magnitude is equivalent to the arc length, or to find another equation to find the arc length.

Arc length

The arc length of a function is the distance of a defined section of a curve defined between two points. For the arc length to be determined, the function and its derivative must be continuous between these two points.

As I did not know how to find the arc length of the parametric curve, I wanted to start by finding the arc length of a normal curve, then converting it to a parametric curve in some way. As doing numerical integration to estimate the area under a curve would divide the curve into small sections to simplify the curve, I would also try and find the arc length through dividing it into small sections.

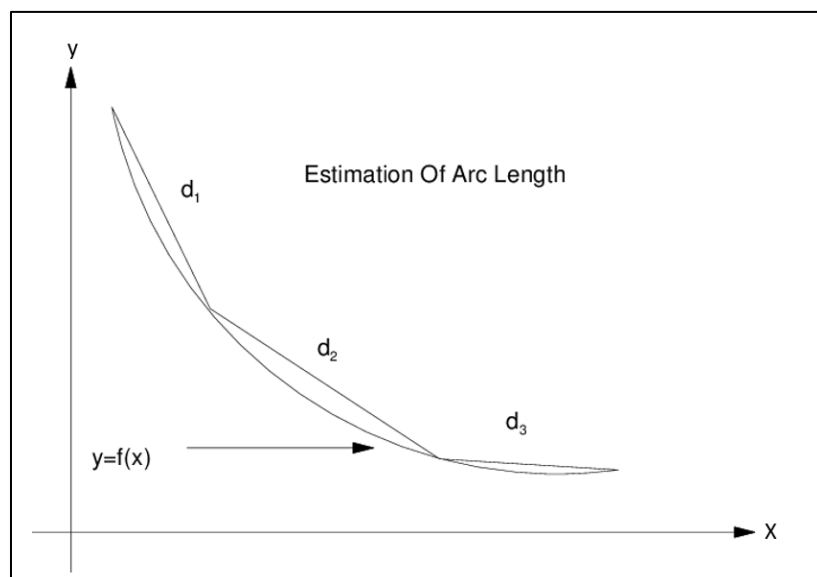


Figure 14: Estimations of the Intervals of the arc length (Sugden and Simmond)

The curve is divided into equal intervals of n with width Δx . denoting the point on the curve at each point by P_i , the curve would then be approximated by connecting these dots with straight lines.

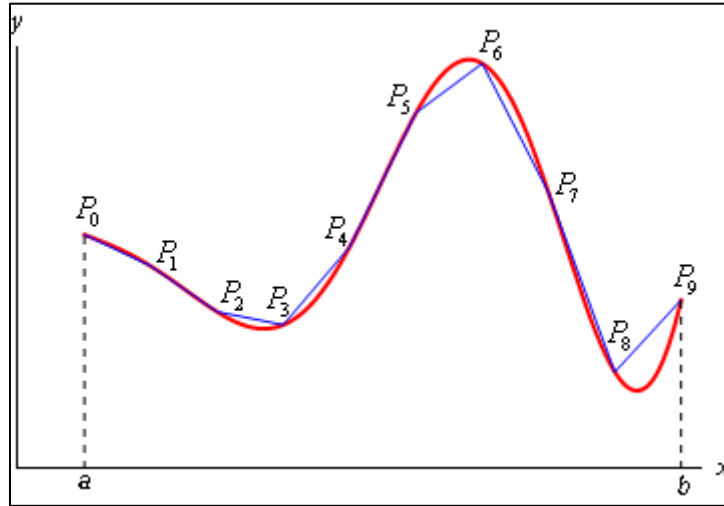


Figure 15: Intervals of arc length with points (Dawkins)

The length of these lines can be summed up using the formula

$$L \approx \sum_{i=1}^n |P_{i-1}P_i| \quad (19) \text{ (Dawkins)}$$

As n tends to infinity, the arc length would become more accurate, so by taking the limit of (19), (19) becomes

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i| \quad (20)$$

The length of the line segments can be computed as follows:

$$|P_{i-1}P_i| = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} \quad (21)$$

After getting stuck on this equation, I found out that I needed to use a way to substitute something into the x or y so that the length of the line segments could be found.

For the change in y , Δy_i can be defined as

$$\Delta y_i = y_i - y_{i-1} = f(x_i) - f(x_{i-1}) \quad (22)$$

Using the Mean Value theorem, which states that if f is continuous over the closed interval $[a,b]$ and differentiable over the open interval (a,b) then there exists a point $c \in (a,b)$ such that the tangent line to the graph of f at c is parallel to the secant line connecting $(a,f(a))$ and $(b,f(b))$. (Libretexts)

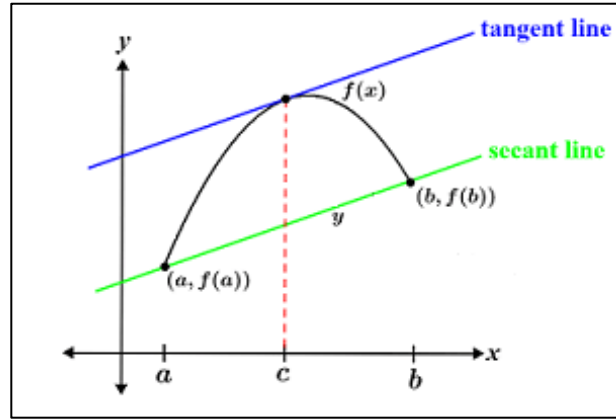


Figure 16: Diagram of Mean Value Theorem (Mean value theorem)

Shown in a formulaic form, it is:

$$f'(c) = \frac{f(b)-f(a)}{(b-a)} \quad (23) \text{ (Libretexts)}$$

On the interval $[x_i, x_{i-1}]$, there is a point x_i^* , which is c , and substituting b for x_{i-1} and a for x_i , the formula becomes

$$f'(x_i^*) = \frac{f(x_i)-f(x_{i-1})}{(x_i - x_{i-1})}$$

$$f(x_i) - f(x_{i-1}) = f'(x_i^*)(x_i - x_{i-1}) \quad (24)$$

Using equation (22) and (24), the change in y can be stated as

$$\Delta y_i = f'(x_i^*)\Delta x \quad (25)$$

Using (21) and (25), the length can be written as

$$|P_{i-1}P_i| = \sqrt{\Delta x^2 + f'(x_i^*)^2\Delta x^2} \quad (26)$$

$$|P_{i-1}P_i| = \sqrt{1 + f'(x_i^*)^2}\Delta x \quad (27)$$

Substituting (27) into (19), we get

$$L = \lim_{n \rightarrow \infty} \sum \sqrt{1 + f'(x_i^*)^2}\Delta x \quad (28)$$

Using the Riemann integral, which is defined by taking the limit of a Riemann sum, which

are used to approximate $\int_a^b f(x)dx$ by using the areas of rectangles or trapezoids

(Oberbroeckling) for approximating the area under a curve, the definite integral of $f(x)$ over the open interval (a,b) can be expressed as

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f'(x_i^*) \Delta x \quad (29) \text{ (Ranguwar)}$$

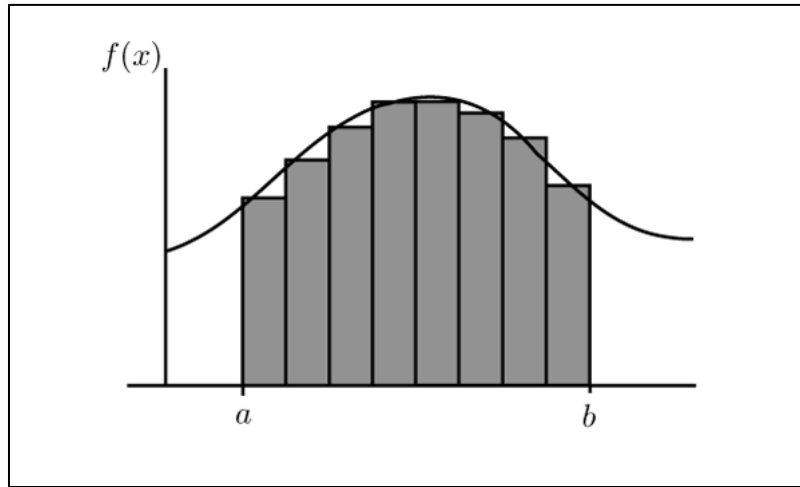


Figure 17: Diagram of Riemann Sum (Riemann_Integration_4.png)

(26) then can be expressed as

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad (30) \text{ (Dawkins)}$$

Similarly, using it for the arc length with respect to y , the formula is

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \quad (31)$$

For parametric equations, we can assume that $\frac{dx}{ds} \geq 0$ for $a \leq s \leq b$, so

$$dx = f'(t) \quad (32)$$

$$dt = \frac{dx}{\frac{dx}{dt}} \quad (33)$$

Using formula (31), (29) becomes

$$L = \int_a^b \sqrt{1 + \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)^2} \frac{dx}{dt} dt \quad (34)$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \frac{dx}{dt} dt \quad (35)$$

Finally, we can factor out the denominator, so (34) becomes

$$L = \int_a^b \frac{1}{\left|\frac{dx}{dt}\right|} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \frac{dx}{dt} dt \quad (36)$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (37) \text{ (Dawkins)}$$

Using this relationship, we can also extend this to the third dimension, z, as the component of the rate of change of would also be the same as the x and y coordinate. As the z coordinate changes in a linear proportion, the rate of change can be expressed as

$$\frac{dz}{dt} = w$$

$$z = \int_a^b w$$

Where w is the rate of change of the z dimension. This can also be extended to the vector function (17) and arc length function (38), and so they are shown below:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \quad (37)$$

$$\vec{h}(t) = \langle \int_a^b \cos\left(\frac{1}{2}qs^2\right), \int_a^b \sin\left(\frac{1}{2}qs^2\right), \int_a^b w \rangle \quad (38)$$

After much research and analysis, I finally got a formula to relate the rate of change of the parametric function to its arc length, I understood more about the relationship between them and the effect that they have on each other.

Final Answer

For function (38), when we derive it, we can get

$$\vec{h}'(t) = \langle \cos\left(\frac{1}{2}qs^2\right), \sin\left(\frac{1}{2}qs^2\right), w \rangle \quad (39)$$

Substituting formula (39) into (37) we can get the arc length of the clothoid curve

$$|\vec{h}'(t)| = \sqrt{\cos\left(\frac{1}{2}qs^2\right)^2 + \sin\left(\frac{1}{2}qs^2\right)^2 + w^2} \quad (39)$$

Then using the formula

$$\sin^2 x + \cos^2 x = 1 \quad (40)$$

(38) becomes

$$\vec{h}'(t) = \sqrt{1 + c^2}$$

So, the arc length over the interval [a, b] is

$$L = \int_a^b \sqrt{1 + c^2} dt$$

$$L = \sqrt{1 + w^2} b - a \quad (40)$$

As the clothoid curve only applies for the roller coaster at one half, the final equation is

$$L_T = 2\sqrt{1 + w^2} (b - a) \quad (41)$$

Reflection

Through this investigation, I have achieved my goal of learning more about rollercoaster loops and the arc length of it. To extend this investigation, I can try to find the arc length of different elements of the rollercoaster track, such as the corkscrew loop and the pretzel loop. In addition, I could use the final formula of the arc length in order to calculate the volume and the total surface area of the rollercoaster loop. I could also use this formula to calculate the velocity of the rollercoaster as it goes through a loop and the time taken for it to finish it.

My calculations were valid for this investigation. Looking at the results, the arc length of a rollercoaster loop is determined by the height, length and the width of the rollercoaster loop,

and also how these values changes over the distance travelled by the rollercoaster as it goes through the loop. I understood more about trigonometry and curvature of circular objects through the discovery of the clothoid curve as the shape of a rollercoaster loop, finding out how different shapes in real life could be found utilising trigonometric functions. After deriving out the arc length formula both for a normal function and a parametrised curve, I learnt more about finding the length of different real-life objects can be done using these formulas, and how architects and engineers use this formula to create structures with rarely seen shapes.

However, there are some limitations to my calculations. There are also other ways of computing a rollercoaster loop, so the model I use may not be suitable for other rollercoasters, which uses physics equations relating to forces and motion to derive a loop out. More methods can be used to find out the arc length instead of formula (36), such as using the Euler's method to estimate the arc length of a curve.

As the clothoid curve can also be used to model different things such as a track transition curve for trains and cars so that they maintain centripetal acceleration and not drive off the road or the track and also be used for Physics in the field of optics through modelling a diffraction pattern, this investigation can be used to find the arc length of these curves and the pattern of a diffraction pattern, so using the clothoid curve, we can understand more about these things.

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